
CONTINUOUS RANDOM VARIABLES

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I Introduction

There are random variables whose set of possible values is either finite or countably infinite. However, there also exist random variables whose set of possible values is uncountable. Two examples are the time that a train arrives at a specified stop and the lifetime of a transistor. Let X be such a random variable. We say that X is a *continuous*[†] random variable if there exists a nonnegative function f , defined for all real $x \in (-\infty, \infty)$, having the property that for any set B of real numbers,[‡]

$$P\{X \in B\} = \int_B f(x) dx \quad (1.1)$$

The function f is called the *probability density function* of the random variable X . (See Figure 1.)

In words, Equation (1.1) states that the probability that X will be in B may be obtained by integrating the probability density function over the set B . Since X must assume some value, f must satisfy

$$1 = P\{X \in (-\infty, \infty)\} = \int_{-\infty}^{\infty} f(x) dx$$

All probability statements about X can be answered in terms of f . For instance, from Equation (1.1), letting $B = [a, b]$, we obtain

$$P\{a \leq X \leq b\} = \int_a^b f(x) dx \quad (1.2)$$

[†]Sometimes called *absolutely continuous*.

[‡]Actually, for technical reasons, Equation (1.1) is true only for the *measurable* sets B , which, fortunately, include all sets of practical interest.

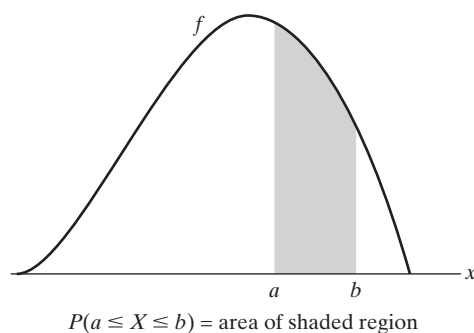


Figure 1 Probability density function f .

If we let $a = b$ in Equation (1.2), we get

$$P\{X = a\} = \int_a^a f(x) dx = 0$$

In words, this equation states that the probability that a continuous random variable will assume any fixed value is zero. Hence, for a continuous random variable,

$$P\{X < a\} = P\{X \leq a\} = F(a) = \int_{-\infty}^a f(x) dx$$

Example 1a

Suppose that X is a continuous random variable whose probability density function is given by

$$f(x) = \begin{cases} C(4x - 2x^2) & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) What is the value of C ?
- (b) Find $P\{X > 1\}$.

Solution (a) Since f is a probability density function, we must have $\int_{-\infty}^{\infty} f(x) dx = 1$, implying that

$$C \int_0^2 (4x - 2x^2) dx = 1$$

or

$$C \left[2x^2 - \frac{2x^3}{3} \right] \bigg|_{x=0}^{x=2} = 1$$

or

$$C = \frac{3}{8}$$

Hence,

$$(b) P\{X > 1\} = \int_1^{\infty} f(x) dx = \frac{3}{8} \int_1^2 (4x - 2x^2) dx = \frac{1}{2}$$

■

Example 1b

The amount of time in hours that a computer functions before breaking down is a continuous random variable with probability density function given by

$$f(x) = \begin{cases} \lambda e^{-x/100} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

What is the probability that

- (a) a computer will function between 50 and 150 hours before breaking down?
- (b) it will function for fewer than 100 hours?

Solution (a) Since

$$1 = \int_{-\infty}^{\infty} f(x) dx = \lambda \int_0^{\infty} e^{-x/100} dx$$

we obtain

$$1 = -\lambda(100)e^{-x/100} \Big|_0^{\infty} = 100\lambda \quad \text{or} \quad \lambda = \frac{1}{100}$$

Hence, the probability that a computer will function between 50 and 150 hours before breaking down is given by

$$\begin{aligned} P\{50 < X < 150\} &= \int_{50}^{150} \frac{1}{100} e^{-x/100} dx = -e^{-x/100} \Big|_{50}^{150} \\ &= e^{-1/2} - e^{-3/2} \approx .383 \end{aligned}$$

(b) Similarly,

$$P\{X < 100\} = \int_0^{100} \frac{1}{100} e^{-x/100} dx = -e^{-x/100} \Big|_0^{100} = 1 - e^{-1} \approx .632$$

In other words, approximately 63.2 percent of the time, a computer will fail before registering 100 hours of use. ■

Example 1c

The lifetime in hours of a certain kind of radio tube is a random variable having a probability density function given by

$$f(x) = \begin{cases} 0 & x \leq 100 \\ \frac{100}{x^2} & x > 100 \end{cases}$$

What is the probability that exactly 2 of 5 such tubes in a radio set will have to be replaced within the first 150 hours of operation? Assume that the events $E_i, i = 1, 2, 3, 4, 5$, that the i th such tube will have to be replaced within this time are independent.

Solution From the statement of the problem, we have

$$\begin{aligned} P(E_i) &= \int_0^{150} f(x) dx \\ &= 100 \int_{100}^{150} x^{-2} dx \\ &= \frac{1}{3} \end{aligned}$$

Hence, from the independence of the events E_i , it follows that the desired probability is

$$\binom{5}{2} \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^3 = \frac{80}{243} \quad \blacksquare$$

The relationship between the cumulative distribution F and the probability density f is expressed by

$$F(a) = P\{X \in (-\infty, a]\} = \int_{-\infty}^a f(x) dx$$

Differentiating both sides of the preceding equation yields

$$\frac{d}{da}F(a) = f(a)$$

That is, the density is the derivative of the cumulative distribution function. A somewhat more intuitive interpretation of the density function may be obtained from Equation (1.2) as follows:

$$P\left\{a - \frac{\varepsilon}{2} \leq X \leq a + \frac{\varepsilon}{2}\right\} = \int_{a-\varepsilon/2}^{a+\varepsilon/2} f(x) dx \approx \varepsilon f(a)$$

when ε is small and when $f(\cdot)$ is continuous at $x = a$. In other words, the probability that X will be contained in an interval of length ε around the point a is approximately $\varepsilon f(a)$. From this result, we see that $f(a)$ is a measure of how likely it is that the random variable will be near a .

**Example
1d**

If X is continuous with distribution function F_X and density function f_X , find the density function of $Y = 2X$.

Solution We will determine f_Y in two ways. The first way is to derive, and then differentiate, the distribution function of Y :

$$\begin{aligned} F_Y(a) &= P\{Y \leq a\} \\ &= P\{2X \leq a\} \\ &= P\{X \leq a/2\} \\ &= F_X(a/2) \end{aligned}$$

Differentiation gives

$$f_Y(a) = \frac{1}{2}f_X(a/2)$$

Another way to determine f_Y is to note that

$$\begin{aligned} \epsilon f_Y(a) &\approx P\left\{a - \frac{\epsilon}{2} \leq Y \leq a + \frac{\epsilon}{2}\right\} \\ &= P\left\{a - \frac{\epsilon}{2} \leq 2X \leq a + \frac{\epsilon}{2}\right\} \\ &= P\left\{\frac{a}{2} - \frac{\epsilon}{4} \leq X \leq \frac{a}{2} + \frac{\epsilon}{4}\right\} \\ &\approx \frac{\epsilon}{2}f_X(a/2) \end{aligned}$$

Dividing through by ϵ gives the same result as before. ■

2 Expectation and Variance of Continuous Random Variables

The expected value of a discrete random variable X is defined by

$$E[X] = \sum_x xP\{X = x\}$$

If X is a continuous random variable having probability density function $f(x)$, then, because

$$f(x) dx \approx P\{x \leq X \leq x + dx\} \quad \text{for } dx \text{ small}$$

it is easy to see that the analogous definition is to define the expected value of X by

$$E[X] = \int_{-\infty}^{\infty} xf(x) dx$$

**Example
2a**

Find $E[X]$ when the density function of X is

$$f(x) = \begin{cases} 2x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Solution

$$\begin{aligned} E[X] &= \int xf(x) dx \\ &= \int_0^1 2x^2 dx \\ &= \frac{2}{3} \end{aligned}$$

■

**Example
2b**

The density function of X is given by

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find $E[e^X]$.

Solution Let $Y = e^X$. We start by determining F_Y , the cumulative distribution function of Y . Now, for $1 \leq x \leq e$,

$$\begin{aligned} F_Y(x) &= P\{Y \leq x\} \\ &= P\{e^X \leq x\} \\ &= P\{X \leq \log(x)\} \\ &= \int_0^{\log(x)} f(y) dy \\ &= \log(x) \end{aligned}$$

By differentiating $F_Y(x)$, we can conclude that the probability density function of Y is given by

$$f_Y(x) = \frac{1}{x} \quad 1 \leq x \leq e$$

Hence,

$$\begin{aligned} E[e^X] &= E[Y] = \int_{-\infty}^{\infty} xf_Y(x) dx \\ &= \int_1^e dx \\ &= e - 1 \end{aligned}$$

■

Although the method employed in Example 2b to compute the expected value of a function of X is always applicable, there is, as in the discrete case, an alternative way of proceeding.

Proposition 2.1 If X is a continuous random variable with probability density function $f(x)$, then, for any real-valued function g ,

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) dx$$

An application of Proposition 2.1 to Example 2b yields

$$\begin{aligned} E[e^X] &= \int_0^1 e^x dx \quad \text{since } f(x) = 1, \quad 0 < x < 1 \\ &= e - 1 \end{aligned}$$

which is in accord with the result obtained in that example.

The proof of Proposition 2.1 is more involved than that of its discrete random variable analog. We will present such a proof under the provision that the random variable $g(X)$ is nonnegative. (The general proof, which follows the argument in the case we present, is indicated in Theoretical Exercises 2 and 3.) We will need the following lemma, which is of independent interest.

Lemma 2.1

For a nonnegative random variable Y ,

$$E[Y] = \int_0^{\infty} P\{Y > y\} dy$$

Proof We present a proof when Y is a continuous random variable with probability density function f_Y . We have

$$\int_0^{\infty} P\{Y > y\} dy = \int_0^{\infty} \int_y^{\infty} f_Y(x) dx dy$$

where we have used the fact that $P\{Y > y\} = \int_y^{\infty} f_Y(x) dx$. Interchanging the order of integration in the preceding equation yields

$$\begin{aligned} \int_0^{\infty} P\{Y > y\} dy &= \int_0^{\infty} \left(\int_0^x dy \right) f_Y(x) dx \\ &= \int_0^{\infty} x f_Y(x) dx \\ &= E[Y] \end{aligned}$$

■

Proof of Proposition 2.1 From Lemma 2.1, for any function g for which $g(x) \geq 0$,

$$\begin{aligned} E[g(X)] &= \int_0^{\infty} P\{g(X) > y\} dy \\ &= \int_0^{\infty} \int_{x:g(x)>y} f(x) dx dy \\ &= \int_{x:g(x)>0} \int_0^{g(x)} dy f(x) dx \\ &= \int_{x:g(x)>0} g(x)f(x) dx \end{aligned}$$

which completes the proof.

**Example
2c**

A stick of length 1 is split at a point U having density function $f(u) = 1, 0 < u < 1$. Determine the expected length of the piece that contains the point $p, 0 \leq p \leq 1$.

Solution Let $L_p(U)$ denote the length of the substick that contains the point p , and note that

$$L_p(U) = \begin{cases} 1 - U & U < p \\ U & U > p \end{cases}$$

(See Figure 2.) Hence, from Proposition 2.1,

$$\begin{aligned} E[L_p(U)] &= \int_0^1 L_p(u) du \\ &= \int_0^p (1 - u) du + \int_p^1 u du \\ &= \frac{1}{2} - \frac{(1 - p)^2}{2} + \frac{1}{2} - \frac{p^2}{2} \\ &= \frac{1}{2} + p(1 - p) \end{aligned}$$

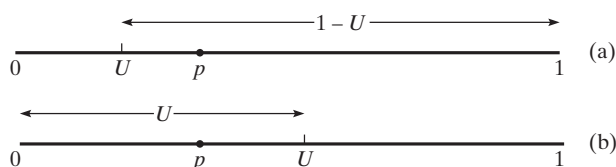


Figure 2 Substick containing point p : (a) $U < p$; (b) $U > p$.

Since $p(1 - p)$ is maximized when $p = \frac{1}{2}$, it is interesting to note that the expected length of the substick containing the point p is maximized when p is the midpoint of the original stick. ■

**Example
2d**

Suppose that if you are s minutes early for an appointment, then you incur the cost cs , and if you are s minutes late, then you incur the cost ks . Suppose also that the travel time from where you presently are to the location of your appointment is a continuous random variable having probability density function f . Determine the time at which you should depart if you want to minimize your expected cost.

Solution Let X denote the travel time. If you leave t minutes before your appointment, then your cost—call it $C_t(X)$ —is given by

$$C_t(X) = \begin{cases} c(t - X) & \text{if } X \leq t \\ k(X - t) & \text{if } X \geq t \end{cases}$$

Therefore,

$$\begin{aligned} E[C_t(X)] &= \int_0^\infty C_t(x)f(x) dx \\ &= \int_0^t c(t - x)f(x) dx + \int_t^\infty k(x - t)f(x) dx \\ &= ct \int_0^t f(x) dx - c \int_0^t xf(x) dx + k \int_t^\infty xf(x) dx - kt \int_t^\infty f(x) dx \end{aligned}$$

The value of t that minimizes $E[C_t(X)]$ can now be obtained by calculus. Differentiation yields

$$\begin{aligned}\frac{d}{dt}E[C_t(X)] &= ct f(t) + cF(t) - ct f(t) - kt f(t) + kt f(t) - k[1 - F(t)] \\ &= (k + c)F(t) - k\end{aligned}$$

Equating the rightmost side to zero shows that the minimal expected cost is obtained when you leave t^* minutes before your appointment, where t^* satisfies

$$F(t^*) = \frac{k}{k + c}$$

We can use Proposition 2.1 to show the following.

Corollary 2.1

If a and b are constants, then

$$E[aX + b] = aE[X] + b$$

The proof of Corollary 2.1 for a continuous random variable X is the same as the one given for a discrete random variable. The only modification is that the sum is replaced by an integral and the probability mass function by a probability density function.

The variance of a continuous random variable is defined exactly as it is for a discrete random variable, namely, if X is a random variable with expected value μ , then the variance of X is defined (for any type of random variable) by

$$\text{Var}(X) = E[(X - \mu)^2]$$

The alternative formula,

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

is established in a manner similar to its counterpart in the discrete case.

Example 2e

Find $\text{Var}(X)$ for X as given in Example 2a.

Solution We first compute $E[X^2]$.

$$\begin{aligned}E[X^2] &= \int_{-\infty}^{\infty} x^2 f(x) dx \\ &= \int_0^1 2x^3 dx \\ &= \frac{1}{2}\end{aligned}$$

Hence, since $E[X] = \frac{2}{3}$, we obtain

$$\text{Var}(X) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}$$

It can be shown that, for constants a and b ,

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

The proof mimics the one given for discrete random variables.

There are several important classes of continuous random variables that appear frequently in applications of probability; the next few sections are devoted to a study of some of them.

3 The Uniform Random Variable

A random variable is said to be *uniformly* distributed over the interval $(0, 1)$ if its probability density function is given by

$$f(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases} \quad (3.1)$$

Note that Equation (3.1) is a density function, since $f(x) \geq 0$ and $\int_{-\infty}^{\infty} f(x) dx = \int_0^1 dx = 1$. Because $f(x) > 0$ only when $x \in (0, 1)$, it follows that X must assume a value in interval $(0, 1)$. Also, since $f(x)$ is constant for $x \in (0, 1)$, X is just as likely to be near any value in $(0, 1)$ as it is to be near any other value. To verify this statement, note that for any $0 < a < b < 1$,

$$P\{a \leq X \leq b\} = \int_a^b f(x) dx = b - a$$

In other words, the probability that X is in any particular subinterval of $(0, 1)$ equals the length of that subinterval.

In general, we say that X is a uniform random variable on the interval (α, β) if the probability density function of X is given by

$$f(x) = \begin{cases} \frac{1}{\beta - \alpha} & \text{if } \alpha < x < \beta \\ 0 & \text{otherwise} \end{cases} \quad (3.2)$$

Since $F(a) = \int_{-\infty}^a f(x) dx$, it follows from Equation (3.2) that the distribution function of a uniform random variable on the interval (α, β) is given by

$$F(a) = \begin{cases} 0 & a \leq \alpha \\ \frac{a - \alpha}{\beta - \alpha} & \alpha < a < \beta \\ 1 & a \geq \beta \end{cases}$$

Figure 3 presents a graph of $f(a)$ and $F(a)$.

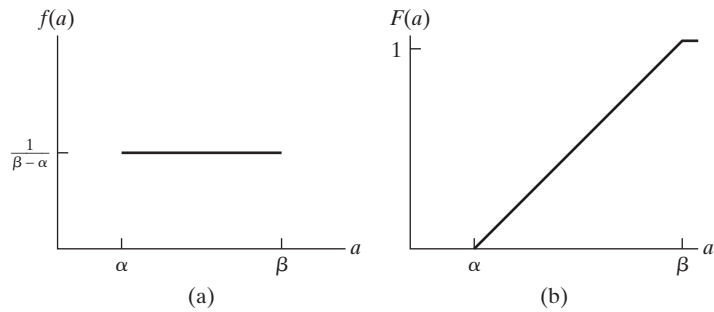


Figure 3 Graph of (a) $f(a)$ and (b) $F(a)$ for a uniform (α, β) random variable.